



Synthesis of starch using CO₂

World-first artificial synthesis of starch from CO₂ outperforms nature.
<https://dgit.in/oct21-71>

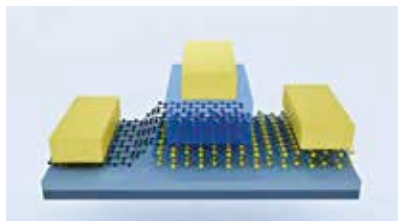


Cryo-Hibernation reality in future?

Supercooling advances human tissue preservation.
<https://dgit.in/oct21-72>

nano electric devices while operating at literally half the voltage of the semiconductors that are available today. The claims are that the current density of this transistor is greater than that of the transistors of the same class.

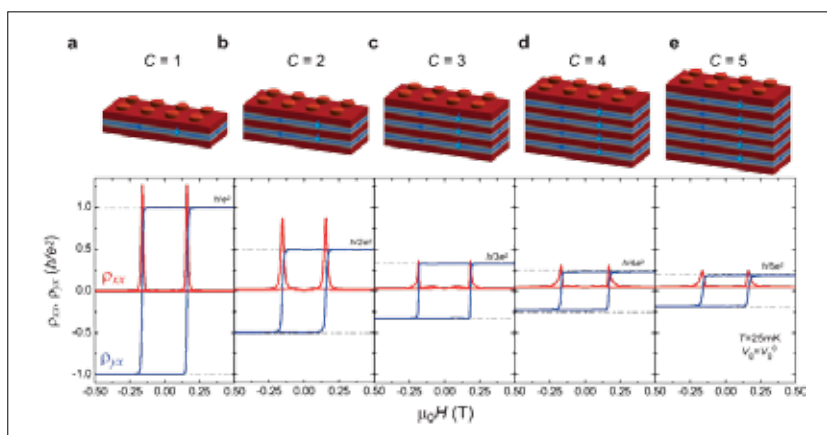
The two materials making up these ‘atom-thin’ transistors are graphene and molybdenum disulphide. When placed over each other, these two materials are just 1 nanometer thick. As stated in the news release, “the unique physical properties of graphene keep electrons “cold” as they get injected from the graphene into the MoS₂ channel. This process is called Dirac-source injection. The electrons are “cold” because they require much less voltage input and, thus, reduced power consumption to operate the transistor.” This results in operational efficiency, much higher than comparable solutions that we have right now.



The black hexagons show graphene, and blue and yellow structures show MoS₂

IT'S THE QUANTUM WAY OR THE HIGHWAY. WHY NOT BOTH?

Sparking a change that could affect the way we use our electrical devices, Penn State University researchers were successful in making what they are calling ‘multilane highways for electrons’. What this could do is, open the world to a whole new era of electronics that are highly efficient and capable of performing tasks of much larger scale and nature, than what they do right now. When we increase the intensity of electrons that are flowing through a metal, into an electronic device, due to space constraints, several electrons do not get enough area to traverse through the metal and are scattered out of the



Looks like Lego, carries electrons

metal, on the edges. This also happens due to the nature of electrons, which causes them to repel each other when they are too close and allows them to travel in two different directions. The name given to this new highway system for electrons was the high Chern number quantum anomalous Hall (QAH) effect. QAH effect is a phenomenon where electrons do not have to encounter any energy loss when they move. The Chern number of a material is the number of pathways or highways that a material can have. If the material has 1 Chern number, it has two highways available for electrons to travel. As per the information published in the study, the QAH effect was realized in materials where the Chern number was 1. These QAH(s) are created in a material known as a topological insulator. What the researchers found in their study was that by changing the thickness of the QAH insulator, they could manipulate the Chern number of the sample. In their experiments, they stacked layers of magnetic and non-magnetic topological insulators and were able to fine-tune the Chern number, which laid the foundations of devices that, in the future, could take advantage of this dissipationless edge current.

THE FUTURE IS BRIGHT

At Georgia Tech, work is underway to understand and further advance focused electron beam-induced pro-

cessing (FEBID) techniques. These techniques, when developed, in the words of Professor Andrei Fedorov, could potentially help the researchers to “activate interaction of the beam with oxygen in the graphene oxide to do etching, or interaction with hydrocarbons on the surface to create carbon deposition.” This would also allow fast and scalable production of 2D and 3D ‘nanostructures and a host of quantum devices, which could come in handy in a variety of applications. The technology essentially allows 3D sculpting on an atomic scale.

A joint venture of SLAC (Stanford Linear Accelerator Center), Stanford and two Illinois Universities, called Center for Quantum Sensing and Quantum Materials, would focus primarily ‘to develop quantum sensors for probing quantum materials’. Under this initiative, three main devices/concepts would be under development and at the centre of focus. The first of these would be a scanning qubit microscope, the second, a spectroscopy instrument that would be taking advantage of entangled electrons, and a third instrument that would be able to use SLAC’s X-ray free-electron laser. The main motive of all these three instruments would be to make way for the development of new quantum materials and develop newer techniques that would look into the behaviours of quantum materials.